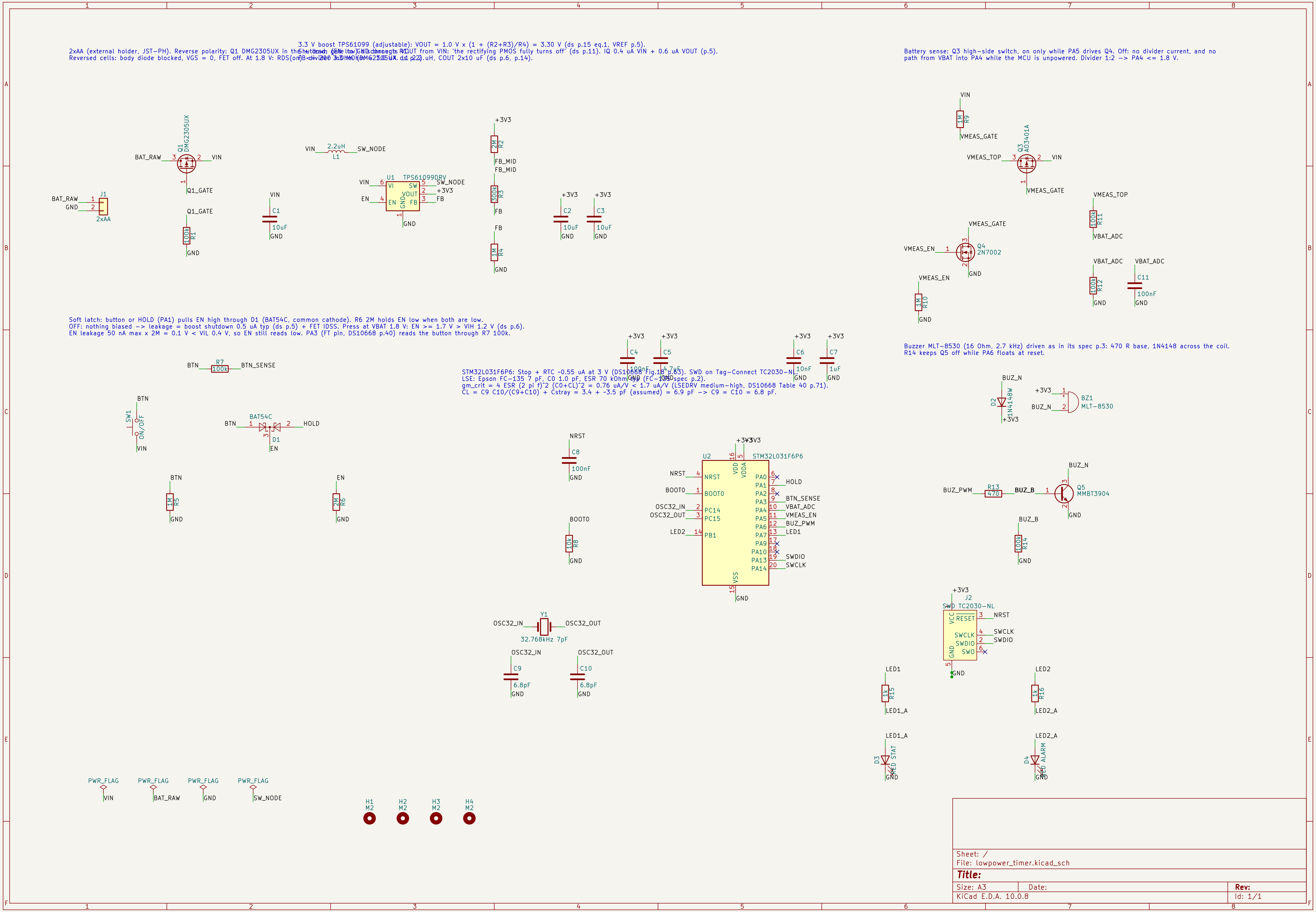

2xAA (external holder, JST−PH). Reverse polarity: Q1 DMG2305UX in the + lead, gate to GND through R1.
Reversed cells: body diode blocked, VGS = 0, FET off. At 1.8 V: RDS(on) <= 200 mOhm (DMG2305UX ds p.2).

3.3 V boost TPS61099 (adjustable): VOUT = 1.0 V x (1 + (R2+R3)/R4) = 3.30 V (ds p.15 eq.1, VREF p.5).
Shutdown (EN low) disconnects VOUT from VIN: 'the rectifying PMOS fully turns off' (ds p.11). IQ 0.4 uA VIN + 0.6 uA VOUT (p.5).
FB divider 3.3 MOhm = 1.0 uA. L1 2.2 uH, COUT 2x10 uF (ds p.6, p.14).

Battery sense: Q3 high−side switch, on only while PA5 drives Q4. Off: no divider current, and no
path from VBAT into PA4 while the MCU is unpowered. Divider 1:2 −> PA4 <= 1.8 V.

Soft latch: button or HOLD (PA1) pulls EN high through D1 (BAT54C, common cathode). R6 2M holds EN low when both are low.
OFF: nothing biased −> leakage = boost shutdown 0.5 uA typ (ds p.5) + FET IDSS. Press at VBAT 1.8 V: EN >= 1.7 V > VIH 1.2 V (ds p.6).
EN leakage 50 nA max x 2M = 0.1 V < VIL 0.4 V, so EN still reads low. PA3 (FT pin, DS10668 p.40) reads the button through R7 100k.

STM32L031F6P6: Stop + RTC ~0.55 uA at 3 V (DS10668 Fig.18 p.63). SWD on Tag−Connect TC2030−NL.
LSE: Epson FC−135 7 pF, C0 1.0 pF, ESR 70 kOhm typ (FC−135 spec p.2).
gm_crit = 4 ESR (2 pi f)^2 (C0+CL)^2 = 0.76 uA/V < 1.7 uA/V (LSEDRV medium−high, DS10668 Table 40 p.71).
CL = C9 C10/(C9+C10) + Cstray = 3.4 + ~3.5 pF (assumed) = 6.9 pF −> C9 = C10 = 6.8 pF.

Buzzer MLT−8530 (16 Ohm, 2.7 kHz) driven as in its spec p.3: 470 R base, 1N4148 across the coil.
R14 keeps Q5 off while PA6 floats at reset.

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